



nextwork.org

VPC Traffic Flow and Security



Courage Dei

✔ Security group (sg-03751bc28fd54ed1e | NextWork Security Group) was created successfully
▶ Details

sg-03751bc28fd54ed1e - NextWork Security Group

Details

Security group name NextWork Security Group	Security group ID sg-03751bc28fd54ed1e	Description A Security Group for the NextWork VPC.	VPC ID vpc-0dd420486cfd46ca7
Owner 352505432441	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules | Outbound rules | Sharing - new | VPC associations - new | Tags

Inbound rules (1)

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-05b2a6e54b425ffa5	IPv4	Custom TCP	TCP	0



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a private network that houses all your resources and it is useful because it prevents other resources from seeing the resources that are not in your vpc

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a route table, security group and network ACL

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the inbound and outbound rules were fascinating

This project took me...

This project took me about 30 minutes



Route tables

Route tables are rule that tell your vpc resources where to go to access what.

Routes tables are needed to make a subnet public because they allow the resouces find it way to a distination

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Q local	-	No	CreateRoute
	Internet Gateway			
	Q igw-0bde3842c018f23b7			



Route destination and target

Routes are defined by their destination and target, which mean the destination is the ip address it can access and the target is the where the gateway that aids the ip's to navigate

The route in my route table that directed internet-bound traffic to my internet gateway had a destination of 0.0.0.0/0 and a target of Internet gateway (igw-0bde3842c018f23b7)

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	-	No	CreateRoute

[Add route](#) [Cancel](#) [Preview](#) [Save changes](#)



Security groups

Security groups are a way to manage the users and resources of a vpc from accessing specific resources on the internet

Inbound vs Outbound rules

Inbound rules are allow users and resources to access contents inside a vpc, I configured an inbound rule that will allow traffic from anywhere on the internet.

Outbound rules are allows our vpc to acces other resouces outside itself be it other vpcs etc By default, my security group's outbound rule is set to allow the vpc access other resouces outside unless i restrict it



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Inbound rules

Outbound rules

Sharing - new

VPC associations - new

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Inbound rules (1)

Search

Name

Security group rule ID

IP version

Type

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Port range

-

sgr-05b2a6e54b425ffa5

IPv4

Custom TCP

TCP

0



Network ACLs

Network ACLs are security rule that handle the broad security of the subnet level by allowing or denying traffic

Security groups vs. network ACLs

The difference between a security group and a network ACL is that security group protect at the instance level while the network ACL does at the subnet level



Default vs Custom Network ACLs

Similar to security groups, network ACLs use inbound and outbound rules

By default, a network ACL's inbound and outbound rules will allow all traffic

In contrast, a custom ACL's inbound and outbound rules are automatically set to deny all traffic

acl-0a9cdc7aead802ed6 / NextWork Network ACL

Details **Inbound rules** Outbound rules Subnet associations Tags

Inbound rules (2) [Edit inbound rules](#)

Filter inbound rules

Rule number	Type	Protocol	Port range	Source	Allow/Deny
100	All traffic	All	All	0.0.0.0/0	Allow
*	All traffic	All	All	0.0.0.0/0	Deny



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